

A rough guide to backyard astronomy

You don't need telescopes, or 20/20 vision to figure out the night sky. All you need is a PC or a smart phone loaded with interactive star maps



High-resolution images of space are available on www.gigagalaxyzoom.org

Sriram Sharma

readersletters@thinkdigit.com

In the year 1600, Galileo was able to look into Jupiter, see its moons, and declare that the earth revolved around the sun instead of the other way around. Today, you can do the same, with a toy telescope that costs 2,000 rupees from a mall.

There are only 6,000 to 9,000 stars visible from Earth to the naked eye even at the best viewing conditions. Even then, by using just a pair of binoculars and software you can see planets, constellations and star clusters, and at least one spiral galaxy on a clear night.

As part of ESA's International Year of Astronomy, three pictures were released last year, the trilogy of pictures zooms further and further in, demonstrating the capabilities of modern astronomy, and is downloadable from their web site at gigagalaxyzoom.org.

The first picture, an 800-million pixel 360-degree panoramic image

reveals the stars and the milky way as seen by the naked eye (from the darkest and best viewing locations in the world.). It is also available as an interactive 3D map where you can rotate, zoom in and out of it in a browser (www.sergebrunier.com/gallerie/pleinciel/360.swf).

The night sky was for mediaeval civilisations, like TV before TV, a galactic PowerPoint presentation that signalled the passing of time and change of season.

"I wanted to show a sky that everyone can relate to — with its constellations, its thousands of stars, with names familiar since childhood, its myths shared by all civilisations," said Serge Brunier, who spent many months compiling this composite picture made out of 1200 photos from both the hemispheres of the Earth.

Stargazing is free, and so is most of the software that supports it. Google may be more renowned for making

maps that look down on Earth from space, but the Sky Map, a free app for Android phones that can construct a map of stars and constellations. It uses a combination of GPS, network signals, in built compass, and other sensors to determine which direction in the sky the phone is facing, pinpointing the position using GPS to work out exactly which direction you are pointing at, in the known universe. You can also search for stars and planets by name, and chase an arrow to pinpoint it in the sky.

No smart phone? In that case, the best software to accompany your laptop or netbook on an outdoor excursion would be Stellarium. This open source planetarium software is the best way for you to discern the stars and planets visible in the night out of your window. It takes in your latitude and longitude coordinates, and then gives you a real-time simulation of every visible object in space. Using this software, you can safely estimate the time, position, and trajectory of planets and stars visible to the naked eye.

Easily viewable planets in city skies include Jupiter, Venus and Mercury. You will also be able to estimate exactly where and when the moon will rise and set.

Since Stellarium works offline, you can load it on a laptop (hopefully with a battery on full charge,) and be fully equipped for a night of star gazing. It comes loaded with illustrations of constellations, a whole bunch of keyword shortcuts for navigation, and is highly customisable. It



Stellarium is a free astronomy software just like Celestia